First-Step Analysis & Absorbing Chains — Complete Formula Reference

Every identity from Ch. 4.2, zero derivations, plus the extra results your problem sets used.

1. The First-Step Principle

- **Idea:** condition on the first move, express what remains in terms of the *same* unknowns, solve the small linear system
- **Expected hitting time** of a target set: $k_i = 1 + \sum_j P_{ij} k_j$, with $k = 0$ on the target
- **Hitting probability** (reach A before B): $h_i = \sum_j P_{ij} h_j$, with $h = 1$ on A and $h = 0$ on B
- **Expected reward/cost:** $k_i = c_i + \sum_j P_{ij} k_j$ (replace the "1" by the cost of the step)
- **Self-loops:** for *probabilities* ignore them (divide the other entries by $1 - P_{ii}$); for *times* multiply by $\frac{1}{1-P_{ii}}$
- **Elimination order:** solve from the state farthest from the target backward

trap: the "1+" pays for the step you take. Forgetting it turns a time equation into a probability equation.

2. Absorbing Chains: Canonical Form

- **Block form:** $P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$, Q = transient→transient, R = transient→absorbing
- **Fundamental matrix:** $N = (I - Q)^{-1} = \sum_{n \geq 0} Q^n$
- N_{ij} = expected visits to transient j starting from i (counts time 0 if $i = j$)
- **Expected absorption times:** $t = N\mathbf{1}$ (row sums of N)
- **Absorption probabilities:** $B = NR$; every row of B sums to 1
- **Ever-visit probability** ($i \neq j$): $P(i \to j) = \dfrac{N_{ij}}{N_{jj}}$
- **Return probability:** $f_{jj} = 1 - \dfrac{1}{N_{jj}}$
- **Variance of absorption time:**

$$\mathrm{Var}(T) = (2N - I)\,t - t \circ t$$

trap: N_{ij} is the expected number of visits, not the probability of visiting: $N_{ij} = P(i \to j) \cdot N_{jj}$.

3. Gambler's Ruin Family

- **Fair walk** on $\{0, \ldots, N\}$, start k: $P(\text{hit } N) = \dfrac{k}{N}$; $\mathbb{E}[T] = k(N - k)$
- **Biased** ($p \neq q$, $r = q/p$): $P(\text{hit } N) = \dfrac{1 - r^k}{1 - r^N}$
- **Biased duration:**

$$D_k = \frac{k}{q - p} - \frac{N}{q - p} \cdot \frac{1 - r^k}{1 - r^N}$$

- **Midpoint start** on $[0, 2k]$: $P(\text{reach top}) = \dfrac{1}{1 + r^k}$
- **Ties** (stay w.p. s): win probabilities unchanged, times $\times \dfrac{1}{1 - s}$
- **Reflecting wall, hit N from 0:** hard wall (move to 1 w.p. 1): N^2; sticky wall (stay w.p. $\frac{1}{2}$): $N(N+1)$. Trick: $g_i = k_i - k_{i+1}$ is an arithmetic progression
- **Bold play:** in an *unfavourable* game bet as much as allowed (Mosteller 37: \$20 → \$40 at $p = \frac{9}{19}$ gives 0.474 boldly vs. 0.11 timidly)
- **Optional stopping:** fair game ⇒ $\mathbb{E}[X_T] = X_0$, but with *overshoot* you need a first-step system (several stopping values)

trap: a "fair game" formula k/N is only valid when the walk lands exactly on the barrier.

4. Pattern Waiting Times & Penney

- **Suffix-state chain:** states = longest suffix that is a prefix of the pattern(s); one first-step equation per state
- **Conway's rule** (fair coin): $\mathbb{E}[T] = \sum 2^k$ over all k for which the length-k prefix equals the length-k suffix
- **Values:** HH = 6, HT = 4, HHT = 8, HTH = 10, HHH = 14
- **Either of two patterns:** HHT or THH → $\mathbb{E}[T] = 6.5$ (shorter than either alone)
- **r successes in a row** (prob. p each): $\mathbb{E}[T] = \dfrac{(1/p)^r - 1}{1 - p}$ (two sixes: 42)
- **Penney's game:** odds that B beats A are $(AA - AB) : (BB - BA)$ with leading numbers $XY = \sum 2^{k-1}$ over overlaps. E.g. $P(\text{THH before HHT}) = \frac{3}{4}$

trap: patterns that overlap with themselves (HHH, HTH) take longer; non-self-overlapping ones (HHT, HT) finish sooner.

5. Random Walks on Graphs

- **Lump by symmetry:** track distance to the target instead of the vertex
- **Commute-time identity:** $h(u \to v) + h(v \to u) = 2|E|\, R_{\text{eff}}(u, v)$
- **n-cycle, antipode:** $(n/2)^2$ (hexagon: 9)
- **Cube, antipode:** 10 (check: $2|E|R_{\text{eff}} = 24 \cdot \frac{5}{6} = 20$, halve by symmetry)
- **Complete graph K_n cover time:** $\sum_{j=1}^{n-1} \dfrac{n-1}{n-j} = (n-1)H_{n-1}$ (K_4: 5.5)
- **Pendant vertex:** triangle + pendant 4 attached to 3: $h(1 \to 4) = 9$, $h(3 \to 4) = 7$

trap: in a random walk on a graph, each step goes to a uniformly random neighbour — the walker cannot stay put (unless the graph has loops).

6. Renewal Recurrences & Tennis

- **Landing probability** (jumps j w.p. p_j): $u_n = \sum_j p_j u_{n-j}$, $u_0 = 1$, with limit $\dfrac{1}{\mu}$ (μ = mean jump). Frog $\pm 1 / \pm 2$: $u_5 = \frac{21}{32}$, limit $\frac{2}{3}$
- **Deuce** (point prob. p, $q = 1 - p$): $P(\text{win}) = \dfrac{p^2}{p^2 + q^2}$; $\mathbb{E}[\text{points}] = \dfrac{2}{p^2 + q^2}$
- **Game from 0–0:**

$$P = p^4(1 + 4q + 10q^2) + 20p^3q^3 \cdot \frac{p^2}{p^2 + q^2}$$

- **Numbers:** $p = 0.6$: deuce 0.692, game 0.736, E[points in game] ≈ 6.48; $p = 0.5$: 6.75

trap: each scoring layer magnifies the stronger player's edge: 60% per point ⇒ 74% per game.

Quick Decision Guide — "Which Tool Do I Need?"

- **"Expected number of steps until . . . "** → $k_i = 1 + \sum P_{ij} k_j$ with $k = 0$ on the target.
- **"Probability of A before B"** → $h_i = \sum P_{ij} h_j$, boundary values 1 and 0; no "1+".
- **Many absorbing exits or many transient states** → canonical form: $N = (I - Q)^{-1}$, $B = NR$, $t = N\mathbf{1}$.
- **Which exit is touched by only one state?** → write that state's equation first, express the others as multiples of it.
- **Coin/dice pattern** → suffix-state chain, or Conway's $\sum 2^k$ rule for a single pattern; Penney odds for pattern races.
- **Symmetric graph or cube/cycle** → lump states by distance to target; or use commute time $2|E|R_{\text{eff}}$.
- **Gambler's ruin** → fair: k/N, $k(N-k)$; biased: use $r = q/p$; ties change only times; bold play in unfair games.
- **Walk can leap over the barrier** → not k/N; set up the full first-step system (optional stopping alone is not enough).
- **Long chain with a hard core** → decompose: (paths avoiding the core) + (prob. of reaching it) × (core value), e.g. deuce.
- **Always sanity-check:** times must be ≥ the shortest path; probabilities must lie in $[0, 1]$ and rows of B must sum to 1.

Numbers Worth Memorizing Cold

- Fair ruin: k/N and $k(N-k)$ (3 of 10: 0.3, 21)
- Patterns: HH 6, HT 4, HHT 8, HTH 10, HHH 14; HHT-or-THH 6.5; THH beats HHT $\frac{3}{4}$
- Two sixes in a row: 42 (formula $\frac{36-1}{5/6}$)
- Hexagon antipode 9; cube antipode 10; K_4 cover 5.5
- Hard wall N^2 vs. sticky wall $N(N+1)$ (e.g. 36 vs. 42 for $N = 6$)
- Bold vs. timid ($p = 0.45$, \$5→\$20): 0.2025 vs. 0.0318
- Frog $\pm 1 / \pm 2$ lands on stone 5: $\frac{21}{32}$ (limit $\frac{2}{3}$)
- Tennis $p = 0.6$: deuce 0.692, game 0.736, ≈ 6.48 points